

E-Commerce Supply Chain Analytics & Inventory Optimization System

A Professional Supply Chain Analytics Study Combining KPI Analysis, Demand Forecasting, Inventory Optimization, and Predictive Risk Modeling

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Tools Used: Python, Pandas, Scikit-learn, Prophet, Tableau / Power BI

Executive Summary

This report presents a comprehensive supply chain analytics study conducted on large-scale e-commerce transaction data. The project explored operational performance, logistics efficiency, demand forecasting, inventory optimization, and predictive inventory risk analysis using modern analytical techniques. The objective of the project was to evaluate how data-driven analytics can support operational decision-making within modern e-commerce supply chains. The analysis integrated descriptive analytics, forecasting models, inventory planning logic, and machine learning methods to identify operational inefficiencies and generate strategic recommendations. The findings demonstrate that customer demand behavior, logistics responsiveness, supplier reliability, and inventory planning are deeply interconnected operational components within large-scale e-commerce systems.

Project Objectives

- Analyze operational supply chain KPIs and fulfillment performance.
- Evaluate demand trends using forecasting techniques.
- Develop inventory optimization strategies using safety stock and reorder point analysis.
- Build predictive machine learning models for inventory risk classification.
- Generate business-oriented operational recommendations.

Methodology Overview

The project followed a structured analytics workflow beginning with data cleaning and exploratory analysis, followed by KPI development, demand forecasting, inventory optimization modeling, and predictive risk classification. Python was used for analytics and machine learning development, while dashboard visualization tools were integrated for business intelligence reporting.

Operational KPI & Forecasting Insights

Operational KPI analysis revealed variability in shipping performance, return behavior, and fulfillment consistency across product categories and locations. Shipping delays emerged as one of the strongest contributors to operational instability and inventory uncertainty.

Demand forecasting analysis identified recurring fluctuations in customer purchasing behavior over time. Trend analysis showed that customer demand exhibits both short-term volatility and longer-term demand movement patterns. The forecasting model demonstrated how predictive analytics can support:

- proactive inventory planning
- warehouse preparation
- resource allocation
- replenishment scheduling
- operational demand visibility

Key Forecasting Interpretation

The forecasting outputs reinforced the importance of transitioning from reactive inventory management toward more proactive, prediction-driven supply chain planning systems.

Inventory Optimization & Machine Learning Findings

Inventory optimization analysis demonstrated that inventory exposure differs substantially across categories due to demand variability and lead-time differences. High-demand categories required larger safety stock buffers and higher reorder points to maintain operational stability.

A Random Forest machine learning model was developed to classify inventory risk exposure using product, pricing, seller, and logistics-related variables. The model achieved an overall accuracy score of 90%; however, detailed evaluation revealed significant class imbalance behavior. The model predominantly classified observations as low risk and failed to effectively identify minority high-risk inventory cases.

Feature Importance Analysis

Feature	Importance
Price	19.2%
Final Price	18.5%
Discount	16.8%
Review Count	13.8%
Rating	9.0%
Seller Rating	8.9%
Shipping Time	4.8%

Feature importance analysis indicated that pricing structure, product popularity, seller reliability, and shipping performance strongly influence inventory behavior within e-commerce supply chains.

Strategic Recommendations & Conclusion

- Implement category-specific inventory management strategies.
- Increase safety stock allocation for volatile demand categories.
- Integrate demand forecasting into replenishment planning workflows.
- Improve logistics coordination in high-delay operational regions.
- Strengthen supplier and seller performance monitoring systems.
- Apply balanced classification techniques such as SMOTE for predictive modeling improvement.
- Expand operational analytics capabilities using additional warehouse and transportation variables.

Conclusion

This project demonstrates how modern supply chain analytics can support operational intelligence and proactive decision-making within e-commerce environments. By integrating KPI analysis, demand forecasting, inventory optimization, and machine learning, the project provides a practical example of how analytics can improve operational visibility and supply chain responsiveness. The findings reinforce the growing importance of predictive analytics, inventory intelligence, and data-driven decision systems within modern logistics and supply chain management.

Future Development Opportunities

Future improvements could include real-time inventory tracking integration, transportation route optimization, warehouse simulation modeling, and advanced deep learning forecasting systems. These additions would further enhance the operational realism and analytical depth of the project.